MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, M1 (HF-1) METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile, 105MM, HE, M1, (HF-1) Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specificatio to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-C-16173	- Corrosion Preventive Compound, Solvent, Cutback Cold Application
MIL-S-50783	- Steel alloy; Special Purpose for Ammunition Components (HF-1).

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC 1315

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2.1 (Cont'd)

MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammuniton
MIL-STD-1169	- Packaging, Packing and Marking for Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

D10551942	- Projectile, 105MM, HE, M1 (HF-1) Metal Parts Assembly
F8837835	- Pallet for 105MM, HE, M1, (HF-1), Projectile Empty or Loaded
IEL-10551942	- Index to Inspection Equipment List for Projectile, 105MM, HE, M1 (HF-1), Metal Parts Assembly

PUBLICATIONS

Frankford Arsenal

Operating Instructions for Use of Detroit Testing Machine for Band Seat Inspection, including Drawing D8593429.

8648877	- Procedures for Ultrasonic Test (UT) Steel Projectile Bodies, 105MM, HE, M1 and 105MM, HE, M1, (HF-1).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

ASTM Method - American Society for Testing and Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

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3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specification.

3.2 Steel Quality.-The item contractor shall establish procedures to provide a forge quality steel when procuring steel in accordance with MIL-S-50783(MU), i.e., the steel shall be free of pipe, porosity and heavy segregation. The manufacturer will provide a certification to cite the specification under which the steel was manufactured and specific inspection techniques used to assure compliance with requirements of the specification and the soundness condition stated above.

3.3. Heat treatment.-Heat treatment applied to any component parts shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.3.1 Hot forge - heat treat method.-Prior to finish machining, projectile bodies shall be subjected to a heat treatment consisting of heating to 1700 F and hold at heat for a minimum time of 1 hour, transfer to a second furnace at a temperature of 1150 F, hold at 1150 F for a minimum time of 1 hour and air cool. Projectile bodies may be allowed to cool outside the furnace from 1700 F to 1150 F as long as the temperature of the projectiles does not drop below 1150 F.

3.4 Mechanical properties.

3.4.1 Body.-Mechanical properties specified on the applicable drawing shall apply throughout the component.

3.4.2 Rotating band.-After application and subsequent removal, the rotating band shall have the following mechanical properties:

Tnesile Strength - 30,000 lbs. per square inch (psi) minimum

Elongation - 15 percent in one inch minimum

3.5 Banding.

3.5.1 Band seat.-Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.5.2 Band blank.-The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification.

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3.5.3 Application.-The rotating band shall be applied after final heat treatment of the body, by carefully pressing the band blank into the band seat so as to make contact with the sidewalls for a minimum of eight-tenths of the sidewall depth. The band shall be applied in such manner as not to distort the body wall. There shall be no visible gap between the rotating band and the sidewalls of the band seat (on the outside surface of the projectile) after the rotating band has been machined to final dimension.

3.5.4 Band seating.-The diametrical clearance between the rotating band and band seat shall not exceed .006 inch when the destructive method of testing isued or .009 inches when the non destructive method of testing is ued.

3.6 Metal Defects

3.6.1 Body.-The body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

3.6.2 Body cavity.-The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.6.3 Visual Standards.-Visual standards for metal defects will be established for each contract after the first months production, however, the establishment of visual standards will not waive the requirement for compliance with 3.6.1 and 3.6.2 (see 6.2).

3.7 Proving Ground Functional Test.-The projectile assembly shall reveal no evidence of failure, when tested at excess and service pressure. Metal parts failure shall be:

a. Permanent deformation greater than .010 inches on either the body diameter or bourrelet diameter of the projectiles.

b. Loss or breakup of metal parts components.

c. Shear of rotating band as evidenced by complete lack of engraving.

3.8 Workmanship.-The requirements for workmanship are as shown on the applicable drawing, in the applicable specification and the following:

3.8.1 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

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3.8.2 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.8.3 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch prior to packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.8.4 Base cover weld.-The weld shall be sound, and the welding operation shall be so controlled that there will be adequate fusion between projectile base and base cover.

3.8.5 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.8.6 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust corrosion or other foreign matter.

3.8.7 Protection of rotating band.-The machined rotating band shall be protected from being nicked, burred and otherwise damaged in subsequent handling prior to shipment.

3.8.8 Compatibility.-Corrosion preventive compound specified in the applicable drawing shall be compatible with the high explosive filler to be used. Information relative to compatibility can be obtained from the responsible technical agency by request through the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

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4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of sixty-five metal parts as follows:

55 Metal Parts Assemblies, complete, including paint
5 Metal Parts Assemblies, complete, except paint
5 Sets of Component Parts, unpainted, unassembled, including rotating band blanks. The bodies shall be completely machined, including the rotating band seat.

Fifty of the above painted metal parts assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Heat treatment lots.-Heat treatment lots shall contain body forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.3.1.3 Projectile lots.-When it has been determined that bodies of a heat-treated lot have complied with the mechanical properties requirement of the applicable drawing, they may be regrouped for assembly into projectile lots, to which will be assigned a projectile lot number in accordance with MIL-STD-1168. The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day, or work week. The projectile lot thus constituted shall be the basis for the selection of samples for the functional test hereinafter prescribed.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual inspection for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Tool Control Dimensions.-Dimensions marked tool control shall be gaged at the beginning of each eight hours production or fraction thereof from each machine in operation and prior to any tooling changes. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. All material produced prior to periodic inspection will be rejected and screened whenever a defective tool control dimension is found.

4.3.2.2 Body, Drawing F10551944, covering a detail of Drawing D10551942

Defect	Method of Inspection
CRITICAL	
1. Metal Defective	Visual (Probe light)

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of rotating band seat	Gage
102. Width of rotating band seat	Gage
103. Location of rotating band seat	Gage
104. Thickness through base	Gage
105. Diameter of charge cavity at forward bourrelet location (before nosing)	Gage
106. Length from base of cavity to inside basic diameter (before nosing)	Gage
107. Depth of scoring	Gage
108. Scoring or knurling missing from rotating band seat	Visual
109. Foreign matter on rotating band seat	Visual
MINOR	
201. Evidence of shot or sand blasting on band seat	Visual

4.3.2.3 Metal Parts Assembly, Drawing D10551942 (Prior to paint).

CRITICAL	
1. Metal Defective	Visual
MAJOR	
101. Runout of pitch diameter of fuze thread with bourrelets and front face	Gage
102. Runout of ogive with bourrelets	Gage
103. Runout of body with bourrelets	Gage
104. Runout of rotating band flat with bourrelets	Gage
105. Runout of boattail with bourrelets	Gage
106. Diameter of bourrelets, min.	Gage
107. Diameter of body	Gage
108. Diameter of rotating band flat	Gage
109. Length from front to forward outside basic diameter	Gage
110. Length from rear face to forward basic diameter on boatail	Gage
111. Length from rear face to rear basic diameter	Gage
112. Variation in wall thickness	Gage
113. Location of rotating band	Gage
114. Gap between sidewall of band seat and rotating band	Gage
115. Radius or chamfer at boatail	Gage
116. Foreign matter in cavity	Visual

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of nose	Gage
202. Diameter of rotating band relief groove	Gage
203. Length from front face to rear of forward bourrelet	Gage
204. Length, total	Gage
205. Location of base cover, min.	Gage
206. Rotating band angle (front)	Gage
207. Width of rotating band	Gage
208. Width of rotating band flat	Gage
209. Width of rotating band relief groove	Gage
210. Relief groove at rear of rotating band angle	Gage
211. Presence and size of undercut front and rear of rotating band (if applicable)	Visual (see 6.2)
212. Radii or chamfers, missing or incorrect (except as otherwise classified)	Visual (see 6.2)
213. Surface finish, improper	Visual (see 6.2)
214. Staking notches missing or incorrect (if applicable)	Visual (see 6.2)
215. Burr	Visual
216. Foreign matter (except as otherwise classified)	Visual

4.3.2.4 Metal Parts Assembly, Drawing D10551942 (after paint)

CRITICAL	
1. Metal defective	Visual (Probe light)
2. Pool of paint in cavity	Visual
MAJOR	
101. Weight	Approved Scale
102. Pitch diameter of fuze thread, max.	Gage
103. Minor diameter of fuze thread, max.	Gage
104. Pitch diameter of fuze thread, min.	Gage
105. Diameter of bourrelets, max.	Gage
106. Length of effective fuze thread, min.	Gage
107. Rotating band damaged	Visual
108. Foreign matter in cavity	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Paint on fuze thread	Visual
202. Paint on rotating band bearing surface	Visual
203. Protective coating or interior surfaces inadequate or damaged	Visual
204. Protective coating on exterior surfaces inadequate or damaged	Visual
205. Corrosive preventive compound missing from fuze threads or indicated surface	Visual
206. Marking missing, incorrect or illegible	Visual
207. Burr	Visual
208. Foreing matter (except as otherwise classified)	Visual
209. Paint not dry	Visual/Tactile

4.3.2.5 Pallet, Drawing F8837835.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Improperly packed	Visual
203. Strapping missing, loose or broken	Manual/Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.

4.3.3.1 Steel quality.-Records of test documentation shall be maintained and made available for government review to provide evidence of compliance with the requirements of 3.2.

4.3.3.2 Mechanical properties. (see 6.7)

4.3.3.2.1 Hardness

4.3.3.2.1.1 Heat treatment lots.-In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.3.1.2) shall be subjected to a Brinnel hardness test or any method of hardness testing approved by the procuring activity.

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4.3.3.2.1.2 Miscellaneous lots.-Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.3.3.2.1.1.

4.3.3.2.2 Tension test.

4.3.3.2.2.1 Heat treatment lots.-Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.3.3.2.2.2 Miscellaneous lots.-Samples for tension test of miscellaneous heat treatment lots shall be selected in accordance with 4.3.3.2.2.1. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.3.3.2.2.3 Rotating band.-One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation and subjected to this test to determine compliance with 3.4.2. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.3.3.3 Band seating.

4.3.3.3.1 Tightness.

4.3.3.3.1.1 Nondestructive method.-Five body assemblies from each 1-hour production per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.5.4 shall be cause for rejection of the assemblies represented by the failing sample.

4.3.3.3.1.2 Destructive method.-When nondestructive equipment is not available, one body assembly from each 8-hour production or fraction thereof shall be subjected to this test to determine compliance with 3.5.4. Failure to comply shall be cause for rejection of the assemblies represented by the failing sample.

4.3.3.3.2 Sidewall contract.-Rotating bands removed as specified under 4.3.3.2.2.4 or 4.3.3.3.1.2 shall be examined to determine compliance with 3.5.3. Failure to comply shall be cause for rejection of assemblies represented by the sample.

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4.3.3.4 Ultrasonic inspection.-Each projectile body shall be subjected to an automatic ultrasonic inspection in accordance with the procedures for ultrasonic test (UT) of steel bodies for 105MM, M1 and 105MM, ML (HF-1) number 8648877. Rejection of any projectile body by ultrasonic testing shall be considered as failure to comply with the requirements of 3.6 and shall be cause for rejection of that projectile body. Projectile bodies that pass the ultrasonic inspection shall have the letter "U" stamped as required by the applicable drawing.

4.3.3.5 Base cover weld.-Five body assemblies from each 1 hours production or fraction thereof per welding machine in operation shall be subjected to this test. Failure of the weld to comply with the requirements of 3.8.4 shall be cause for rejection of the production represented by the sample and shall be back-screened 100 percent.

4.3.3.6 Cleaning and application of protective coating.-The sample size and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cuase for taking the action recommended therein.

4.3.3.6.1 Salt spray.-The sample size and acceptance criteria for salt spray testing shall be in accordance with the applicable specification. Failure to comply with the requirements of the applicable drawing shall be cause for taking the action recommended by the applicable specification.

4.3.4 Proving ground functional test.-The following proving ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

TABLE I

Phase	Quantity
Excess pressure	30
Service pressure	20

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4.3.4.2 Production lot sample.-The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Quantity
Excess pressure	10
Service pressure	10

After three consecutive production lots have been ballistically tested and accepted, ballistic testing may be discontinued, however, implementation of ballistic testing can be instituted whenever directed by procuring contracting officer.

4.3.4.3 Excess pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.7 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.4.4 Service pressure.-Samples provided in accordance with Tables I and II will be subjected to this test. Failure of any projectile assembly to comply with 3.7 shall be cause for rejection of the initial production sample or the production lot represented by the sample.

4.3.5 Inspection equipment.-Index of Inspection Equipment List No. IEL-10551942 identifies the inspection equipment required to perform the examination and tests prescribed herein. Reference is made to the IEL for definition of coding format.

4.4 Test methods and procedures.

4.4.1 Mechanical properties. (See 6.7)

4.4.1.1 Hardness.-After a final heat treatment or stress relief, but prior to finish machining, a Brinnel hardness test shall be performed on the outside surface at the band seat location or at any other area on the body deemed necessary for process control. The equipment and procedures for performing this test are prescribed by ASTM-E-10 Method, except that a minimum dwell time shall be 5 seconds when a Brinnel hardness test is used.

4.4.1.2 Tension test.

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4.4.1.2.1 Body.-From the samples provided by 4.3.3.2.2.1 three tension test specimens conforming to the largest obtainable standard "R" type specimen, as prescribed by ASTM Method E-8, shall be taken. Two longitudinally specimens from opposite sides of the body at midwall, in such a manner that the center of the gage length of the machined test bar will fall in line coincident with the center of the rotating band seat and one transverse specimen from the base of the body. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements and with the approval of the procuring agency, the Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements. The criteria for performing this test shall be prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.4.1.2.2 Rotating band.-The samples provided by 4.3.3.2.2.3 shall have the rotating band removed and carefully flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank may be machined so that all serration marks from the band seat are removed, or at the option of the contractor, the specimen blank may be tested without removing the serration marks. A test specimen having one inch gage length and a one-half gage width, and grip ends of sufficient length to accomodate the tensile machine shall be removed from the blank and tested in accordance with the procedures prescribed by ASTM Method E-8.

4.4.2 Band seating.

4.4.2.1 Tightness.

4.4.2.1.1 Non-destructive method.-The samples provided by 4.3.3.3.1.1 shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 6.3).

4.4.2.1.2 Destructive Method.- The body assembly selected in accordance with 4.3.3.3.1.2 should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical protion of their length nearthe front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

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4.4.2.2 Sidewall contract.-The samples provided in accordance with 4.3.3.2.2.4 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch froward and rear of rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other half shall be sectioned into two parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contract to determine compliance with the requirements.

4.4.3 Ultrasonic inspection.-The projectile body shall be examined using ultrasonic equipment of the type and model approved by Frankford Arsenal, SMUFA-Q3200. Procedures for operation of the unit shall be as prescribed by procedures number 8648877. Prior to the start of production, the contractor shall submit a copy of his proposed procedures for complying with the requirement of the manual to Commanding Officer, Frankford Arsenal, Philadelphia, Pa. 19137, ATTN: SMUFA-Q3200, for written approval. After approval, no change in procedure shall be made without prior approval of the above organization.

4.4.4 Base cover weld.-The test of the base cover weld shall be performed as follows: visually inspect the base cover and select the area that appears to have the weakest weld; insert a 5/8" chisel in this area and apply force by use of a one pound steel hammer and an approximate one foot stroke. If the weld breaks loose after 2 strokes, the body assembly will be classed defective. If the weld is found to be secure, the burr made by the chisel may be knocked back into position and accepted.

4.4.5 Protective coating.-The procedures for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and criteria for testing as specified in the applicable specification.

4.4.5.1 Salt spray.-The salt spray test prescribed by the applicable drawing shall be performed utilizing the equipment and procedures specified by ASTM Method B-117.

4.4.6 Proving ground functional test.

4.4.6.1 Excess pressure.-The projectiles, supplied in accordance with 4.3.4, inert loaded and dummy fuzed, will be fired for recovery, in a gun for which the projectile is standard, the tube of which will have a minimum 50 percent life remaining, based on current gun tube condemnation criteria. A propellant charge adjusted to produce 109 to 117 percent of the upper pressure limit of the propellant will be used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile. An attempt will be made to recover at least 60 percent of the projectiles fired in this phase for examination to determine compliance with 3.7.

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4.4.6.2 Service pressure.-The projectiles supplied in accordance with 4.3.4 inert loaded and dummy fuzed will be fired alternately with projectiles from a reference lot for comparison. The projectile shall be fired in a gun for which the round is standard, the tube of which shall have a minimum remaining service life of 50 percent, based upon current gun tube condemnation criteria. The projectile will be fired for accuracy at a range selected by the proof director. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile, to determine compliance with 3.7. Velocity and chamber pressure will be recorded for each round fired.

5. PREPARATION FOR DELIVERY

5.1 Level C.-Minimum military pack.

5.1.1 Level C.-Projectile metal parts assemblies shall be palletized in accordance with Drawing F8837835.

5.2 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with MIL-STD-1169.

5.3. Data Card.-Data card information shall be as specified in MIL-STD-1167

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards to be established at contractors plant in compliance with 3.6.3.

6.3 Band seating.-The operating procedures for use of Detroit Testing Machine as prepared by the Quality Engineering Branch, U.S. Army, Frankford Arsenal, SMUFA-Q3200.

6.4 The following materials complying with specification MIL-C-16173D, Grade 2, have been found to be compatible with composition B and TNT.

a. Cosmoline No. 1102 - E.F. Houghton Co., Phila., Pa.
b. Nox-Rust No. 207, Grade 2 - Daubert Chemical Co., Chicago, Ill.
c. Petrotect 2 - Pennsylvania Refining Co., Butler, Pa.

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6.5 Heat treatment is understood to be any thermal process which brings about a modification in the mechanical and/or metallurgical properties of the metal involved.

6.6 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.7 Alternative mechanical properties testing.-Alternative procedures for mechanical properties testing may be used by the contractor when such testing and procedures provide as a minimum the quality assurance specified herein. Prior to applying the alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to or bettwer than the quality assurance provision specified herein.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project No. 1315-A223

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